

Three-Party System Game

Instructor Guide

Quick Start Checklist

1. Open the game at <https://cellis212.github.io/three-party-game/> (no install), or run `app.R` in RStudio (Section 2)
2. Create a new game: 6 hospitals, 6 insurers, 8 rounds, 1,000 population
3. Project the **Public Display** page; enter decisions on the **Instructor Control** page
4. Prepare blank notecards and pens for each team
5. Confirm defaults: EMTALA off, DSH 0%, Medicare 120%, young cost \$1,000, old cost \$1,600, illness 30%/70%

1 Class Timeline (75–90 minutes)

Phase	Time	What Happens
Pre-class setup	5 min before	Launch app, create game, prepare notecards
Game introduction	10 min	Explain roles, goals, notecard process
Round 1: Baseline	10 min	Price discovery (Medicare 120%)
Round 2: Adjustment	8 min	Students refine strategies
Round 3: Medicare cut	10 min	Medicare drops to 90%, cost-shifting
Round 4: <i>If time</i>	8 min	Another round at Medicare 90%
Round 5: EMTALA	10 min	EMTALA enacted (no DSH), charity hazard
Round 6: <i>If time</i>	8 min	Another EMTALA round
Round 7: DSH payments	10 min	DSH at 70%, government partial fix
Round 8: <i>If time</i>	8 min	Final round, all policies active
Post-game debrief	10 min	Discussion questions

Rounds 4, 6, and 8 are optional. The five required rounds (1, 2, 3, 5, 7) take about 60 minutes.

2 Pre-Class Setup

Step 1: Launch the App. Use one of these:

- **Website (no install):** open <https://cellis212.github.io/three-party-game/>. The first load takes a minute while R loads in the browser. The game lives in that one browser tab: if you reload or close the tab, the game is gone. Each tab is a separate game, so run the whole class from one tab.
- **RStudio:** open `app.R` and click **Run App**. Click “Open in Browser” for the full interface. The game is saved in `data/game.state.db`.

Step 2: Create the Game. In the “Game Setup” box: 6 hospitals, 6 insurers, 8 rounds, 1,000 population. Click “Create New Game.”

Step 3: Open the Display. Project the **Public Display** page (in the left sidebar). Enter decisions on the **Instructor Control** page.

- **Website:** switch between the two pages in the same tab. During negotiations, click “Pop Out Negotiation Board” and show that window on the projector (allow pop-ups for the site).
- **RStudio:** you can also open the same address in a second browser tab, keep it on Public Display, and project it.

Step 4: Prepare Notecards. Distribute blank notecards and pens to each team.

3 Game Introduction — Read to Students

Today we’re going to play a market simulation game. You’ll experience firsthand how the healthcare system’s three major players—hospitals, insurers, and patients—interact to determine prices, coverage, and who gets care.

Hospital teams: You run a hospital. Your main decision is setting a *negotiated rate* with each insurer. A rate of 1.0 means the insurer pays exactly your treatment cost. A rate of 1.2 means they pay 20% above cost.

Insurer teams: You run an insurance company. You set a *premium* and choose your *network*—which hospitals’ contracts to accept.

Patients are simulated by the computer based on premium prices and network proximity.

Key numbers:

- Treating a young patient costs **\$1,000**. Older patient: **\$1,600**.
- Young patients: **30%** chance of getting sick. Older: **70%**.
- Medicare reimburses at **120% of cost**—hospitals profit on Medicare.
- If uninsured and sick, **you are denied care**. No treatment.
- **1,000 people** in this market.
- **Contracting costs:** First contract is free. Each additional costs **\$2,000**. (Example: 3 contracts = \$4,000.)

3.1 Notecard Process (Each Round)

1. **Negotiation (5 min):** Hospital–insurer pairs negotiate face-to-face. When a pair agrees on a rate, they write it on a **notecard** with the rate, both team names, and both signatures. Bring to instructor.
2. **Premium decisions (2 min):** Insurers write their premium on a notecard with their team name and hand it to the instructor.
3. **Results:** Instructor enters everything and clicks “Calculate Round Results.”

3.2 Demo Round — Show Students How It Works

Before the first real round, walk students through a quick example:

Read to Students

Let me show you how this works with a quick demo. I’m going to enter some example decisions so you can see what happens.

Enter into the app:

- **Hospital rates:** Each hospital contracts with **only 1 insurer** at a rate of **1.0** (Hospital 1 → Insurer 1, Hospital 2 → Insurer 2, etc.). No other contracts.
- **Insurer premiums:** All insurers set a premium of **\$400**.

Click “Calculate Round Results” and walk students through the display:

Read to Students

Each hospital contracted with exactly one insurer at a rate of 1.0—the insurer pays exactly what it costs to treat a patient, no markup. Every insurer charged \$400. Since each hospital only has one contract, contracting cost is \$0—the first contract is free. Let’s look at the results.

Point out on the display:

- **Hospital revenue:** Break even on commercial patients (rate = 1.0). Medicare at 120% is where profit comes from.
- **Insurer revenue:** \$400 per enrollee, minus claims at 1.0 rate. Expected claims = $\$1,000 \times 1.0 \times 0.30 = \300 , so insurers should be profitable.
- **Uninsured:** Denied care—don’t appear in hospital volumes.
- **Contracting costs:** \$0 for everyone (one contract each).

Read to Students

Now you can see how the pieces fit together. In the real rounds, you’ll negotiate your own rates and set your own premiums. Can you do better than breaking even?

Click “Reset Current Game” and create a new game to start fresh for Round 1.

4 Round 1: Free Market Baseline

Read to Students

This is a free market. No regulations. If you don’t have insurance and get sick, **you are denied care**. Medicare is at 120% of cost. You have 5 minutes to negotiate rates. Go!

What to do: Start the negotiation timer. Collect notecards and enter rates. After 5 min, give insurers 2 min for premiums. Enter premiums, click “Calculate Round Results.”

What to look for:

- Break-even premium at a 1.2 rate is \$360 ($= \$1,000 \times 1.2 \times 0.30$). Below this = losses.
- Uninsured patients are denied care—no hospital costs.
- At 120% Medicare, hospitals earn 20% margin on older patients.

5 Round 2: Adjustment

Read to Students

You’ve seen Round 1 results. Renegotiate. Hospitals: which insurers gave you volume? Insurers: check whether your premiums covered your claims. If you’re paying out more than you’re collecting, raise prices or narrow your network.

Click “Advance to Next Round.” Same notecard process.

Teaching Note: What You Should Have Seen (Rounds 1–2, Baseline)

- **Hospitals** that contracted with multiple insurers attracted more patients but paid higher contracting costs. This mirrors real-world network participation decisions—hospitals weigh administrative burden against patient volume.
- **Insurers** that priced below expected claims learned about underpricing risk. New insurers entering real markets often underprice to attract members, then face massive losses (this happened repeatedly during the early ACA exchanges).
- **Uninsured patients** are denied care. This was the U.S. reality before EMTALA (1986)—hospitals had no legal obligation to treat patients who couldn't pay.
- **Medicare is profitable** at 120%. This is generous compared to reality—today Medicare reimburses at roughly 87% of cost.

6 Round 3: Medicare Cut to 90%

Read to Students — Policy Change

The U.S. government is in a **budget crisis**. To save costs, they have decided to change Medicare's reimbursement to be **90% of the total cost** instead of 120% of the total cost of providing care. That's a huge swing—hospitals were making 20% profit on every Medicare patient, and now they'll be losing 10% on each one. How will you make up the shortfall?

App: Move “Medicare reimbursement” slider from 120 to 90. Advance round.

Teaching point — Cost-shifting:

- At 120%: $\$1,600 \times 0.20 = \320 profit per Medicare patient
- At 90%: $\$1,600 \times 0.10 = \160 loss per Medicare patient
- Swing: \$480 per patient. Hospitals must raise commercial rates to compensate.

7 Round 4: *If Time* — Another Medicare Round

Same rules (Medicare 90%). Let students adjust further. *Skip if short on time.*

Teaching Note: What You Should Have Seen (Rounds 3–4, Medicare Cut)

- **Hospitals raised commercial rates.** When Medicare stopped being profitable, hospitals cost-shifted to insurers. In real life, hospitals offset ~\$100B in annual Medicare underpayment by charging commercial insurers 150–250% of Medicare rates.
- **Insurers raised premiums.** Higher hospital rates flowed through to consumers. This is exactly what happens when hospital consolidation or rate increases occur in real markets.
- **Some consumers dropped insurance.** Higher premiums pushed price-sensitive consumers out—the premium spiral in miniature.
- **The government “saved” on Medicare but created costs elsewhere.** The budget savings didn't make healthcare cheaper—they moved the cost from taxpayers to commercially insured patients. In an interconnected system, squeezing one payer inflates costs for others.

8 Round 5: EMTALA — The Charity Hazard

Read to Students — Policy Change

Uninsured patients are being denied lifesaving care because they cannot pay. That's not good! The government decides to step in and pass the **Emergency Medical Treatment and Labor Act** (also known as **EMTALA**). Under EMTALA, hospitals are required to treat patients regardless of their ability to pay. Now, uninsured patients get treated but pay nothing. This is a form of **charity care**. How should this impact hospitals? And should this change the value of insurance?

App: Check “EMTALA protections.” Make sure DSH stays at **0%** (manually reset if the app auto-adjusts). Medicare stays at 90%.

Teaching point — Charity hazard:

1. More uninsured—free care makes skipping insurance rational
2. Charity care costs—hospitals treat at full cost, receive \$0
3. Cost-shifting pressure—hospitals must raise commercial rates further

9 Round 6: *If Time* — Another EMTALA Round

Same rules (EMTALA on, no DSH, Medicare 90%). *Skip if short on time.*

Teaching Note: What You Should Have Seen (Rounds 5–6, EMTALA)

- **Uninsured rate jumped.** The *charity hazard*: being uninsured became a viable strategy since hospitals must now treat you for free. In real life, EMTALA passed in 1986. Today ~27 million Americans are uninsured, many relying on ERs as their safety net.
- **Hospitals took massive losses on charity care.** Treating patients at full cost with \$0 revenue is an *unfunded mandate*. U.S. hospitals provided ~\$42 billion in uncompensated care in 2020.
- **Hospitals cost-shifted again.** Facing charity care losses on top of the Medicare cut, hospitals raised commercial rates further. But at some point premiums get so high that more consumers drop insurance, creating more charity care—the *death spiral*.
- **Insurers faced a shrinking market.** Fewer potential customers as the uninsured pool grew. This dynamic contributed to real-world insurer exits from rural and low-income markets.

10 Round 7: DSH Payments — The Government's Partial Fix

Read to Students — Policy Change

The government notices that both hospitals and insurers are struggling. That's not good for anyone! They've decided to step in and partially reimburse the cost of uninsured patients by creating what's known as **Disproportionate Share Hospital payments** (also known as **DSH payments**). What do these do? Hospitals serving uninsured patients now receive **70% of the cost** of charity care from the government. Hospitals now only lose 30 cents on every dollar of charity care.

App: Move “DSH reimbursement” slider to 70. EMTALA stays checked. Medicare stays at 90.

Teaching point: DSH at 70% means \$700 revenue per young charity care patient instead of \$0. But hospitals still lose \$300 per patient. And taxpayers are now funding the program.

11 Round 8: *If Time* — Final Round

All policies remain: EMTALA, DSH 70%, Medicare 90%. Final adjustment round.

Teaching Note: What You Should Have Seen (Rounds 7–8, DSH Payments)

- **Hospital finances improved but didn't fully recover.** At 70% DSH, hospitals receive \$700 per young charity care patient instead of \$0, but still lose \$300. Combined with the Medicare shortfall, hospitals are squeezed from two directions. In real life, DSH payments total over \$40B annually in federal spending.
- **Uninsured rate stayed high.** DSH pays hospitals, not consumers—it does nothing to make insurance more affordable. The charity hazard persists. DSH was never intended to reduce the uninsured rate; that was the goal of later policies like the ACA.
- **Taxpayers are now funding the system.** The government cut Medicare to save money (R3), then spent money on DSH to fix EMTALA's side effects (R7). Each intervention creates side effects requiring further intervention.
- **No one is fully satisfied.** Hospitals make thin margins. Insurers serve a smaller market. Consumers pay higher premiums. Taxpayers fund DSH. Uninsured patients lack preventive care. This is remarkably close to the actual U.S. healthcare system—a patchwork where every stakeholder has legitimate grievances.

12 Post-Game Debrief

12.1 Discussion Questions

1. What happened to premiums over the rounds? Why didn't competition drive them to zero?
2. When Medicare was cut from 120% to 90%, what did hospitals do? Who paid?
3. Before EMTALA, uninsured patients were denied care. After EMTALA, what happened to the uninsured rate? Why?
4. Did DSH fully solve EMTALA's problem? What did it fix? What did it leave broken?
5. Who pays for DSH? Is it free money?
6. Medicare reimburses at about 87% in real life. What does our game say about what that means for everyone else?
7. Could hospitals, insurers, and patients all be made better off simultaneously?

12.2 Real-World Connections

Game Concept	Real-World Parallel
EMTALA checkbox	EMTALA (1986)—hospitals must stabilize all ER patients
DSH payments	Medicaid/Medicare DSH—\$40B+ annually
Medicare 120% → 90%	Medicare reimburses ~87% of hospital costs nationally
Denied care (pre-EMTALA)	Pre-1986 reality—hospitals could turn away patients
Cost-shifting	Commercial insurers pay ~150–250% of Medicare rates
Contracting costs	Real administrative costs of network participation

13 Quick Reference

13.1 Break-Even Premium

$$\text{Break-even} = \text{base cost} \times \text{rate} \times \text{illness probability}$$

Scenario	Base Cost	Rate	Illness Prob	Break-Even
Default	\$1,000	1.2	30%	\$360
Rate = 1.0	\$1,000	1.0	30%	\$300
Rate = 1.5	\$1,000	1.5	30%	\$450
After cost-shift	\$1,000	1.4	30%	\$420

13.2 DSH Math (Charity Care)

$$\text{DSH revenue per patient} = \text{base cost} \times \text{DSH}\%$$

At DSH 70%: hospital receives \$700 per young patient, loses \$300.

13.3 Policy Lever Summary

Control	Default	Change	Round
Medicare reimbursement	120%	→ 90%	R3
EMTALA	Off	→ On	R5
DSH reimbursement	0%	→ 70%	R7

13.4 Default Parameters

Parameter	Value
Hospitals / Insurers	6 each
Rounds	8 (5 required + 3 optional)
Population	1,000 (70% young, 30% old)
Contracting cost	First free, \$2,000 each additional
Young / Old base cost	\$1,000 / \$1,600
Illness rates	30% young, 70% old
Medicare	120% of cost
DSH / EMTALA	0% / Off

13.5 Round Summary

Round	Policy State	Key Lesson
R1	Baseline (Medicare 120%)	Price discovery
R2	Adjustment	Strategy refinement
R3	Medicare → 90%	Cost-shifting
R4*	(same)	Further adjustment
R5	+EMTALA (no DSH)	Charity hazard
R6*	(same)	Further adjustment
R7	+DSH 70%	Government partial fix
R8*	(same)	Full policy regime

*Optional “if time” rounds

14 Troubleshooting

Calculate button doesn't work

Check that some decisions have been entered. You can calculate even if not all teams submitted—missing teams get defaults.

Display not updating

The Public Display refreshes every few seconds. On the website, do not reload the page (that ends the game); switch to Instructor Control and back. In RStudio, you can reload the display tab.

Pop-out board does not open

Allow pop-ups for the site in the browser, then click “Pop Out Negotiation Board” again.

Wrong number entered

Re-enter the correct value and recalculate.

Want to redo a round

Click “Calculate Round Results” again—it overwrites. Re-enter decisions first.

Want to restart

Click “Reset Current Game” (red button). Clears all data.

App crashed

RStudio: stop the R process and click Run App again; the game state is saved in the database.

Website: the game cannot be recovered after a reload, so download the results CSVs (Instructor Control) after each round if you want a backup.

DSH slider moved on its own

The app may auto-adjust DSH when EMTALA is toggled. Manually set it to the value you want.